Academic Portfolio

Postdoctoral Fellowship Application
Faculty of Arts, Shenzhen University — 2026
Supervisor: Professor Liu Kun

Dr Ignasi Solé Piñas

Ph.D. Musicology, University of Aberdeen (2025)
Empirical Performance Studies · Historical Recordings · Beethoven

London, W13 0FD, United Kingdom
ignasiphd@gmail.com · +44 7927 946 976
linkedin.com/in/isole

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1. Cover Letter

Dr Ignasi Solé Piñas
London, W13 0FD, United Kingdom
ignasiphd@gmail.com · +44 7927 946 976
linkedin.com/in/isole

28 April 2026

Professor Liu Kun
Faculty of Arts
Shenzhen University
Shenzhen, China

Dear Professor Liu,

I am writing to apply for a postdoctoral fellowship in your team at the Faculty of Arts, Shenzhen University, advertised for the 2026 cohort in musicology and art theory. I completed my Ph.D. in Musicology at the University of Aberdeen in 2025, with a thesis titled *Tempo and Performance Style in Recordings of Beethoven's Cello Sonatas, 1930–2012*. I am now seeking the opportunity to continue and extend that research in a research-active environment that brings musicology, music education, history and sociocultural studies into close dialogue — a combination that, as I understand it, defines the intellectual character of your team.

My doctoral research provides the first systematic empirical analysis of stylistic change across the complete recorded history of Beethoven's five sonatas for piano and cello. From a corpus of more than one hundred movement-level recordings spanning eight decades I derived bar-level tempo data with millisecond resolution, using a manual measurement protocol I developed specifically for polyphonic chamber recordings on which automated beat-detection tools systematically fail. The work yielded three substantive findings: a quantified gap between Czerny's, Moscheles's and Kolisch's historical tempo indications and recorded practice; the persistence of three coexisting tempo traditions stable across eight decades, identified through k-means clustering; and a measurable decline of audible portamento across the twentieth century. The dataset, annotated scores and Python/MATLAB analysis code are openly published, and five papers based on the thesis have been submitted to arXiv between January 2025 and April 2026.

I see clear points of contact with the work of your team. The Czerny/Moscheles/Kolisch paper is in essence an exercise in music criticism and historiography: it reads nineteenth- and twentieth-century editorial authority against the evidence of the twentieth-century performing tradition. The k-means analysis advances what I have called an *ecological model of stylistic change*, in which coexisting performance traditions shift in relative density rather than a single style evolving — a sociocultural rather than purely technical claim about how performing communities transmit and revise practice. And my long teaching career has shaped my interest in how empirical research on performance can inform music education at every level, from the curricula I currently design at primary and secondary level to the historically informed performance debates that reach into university and conservatoire teaching.

Although my professional life over the past five years has been based in school music leadership in

London, where I currently serve as Director of Music at Notting Hill and Ealing High School, this employment has been the practical means by which I self-funded my doctoral research. Throughout this period I have maintained an active research programme, producing the 346-page thesis and the five arXiv papers alongside my teaching duties. The Shenzhen University post would be my first full-time research position, and I am ready to make that transition: my research pipeline is already in motion, and my priorities over the fellowship would be a monograph proposal, an extension of the methodology to Beethoven's violin sonatas and selected string quartets, and the development of a hybrid manual-and-automated audio-analysis pipeline that could in principle be extended to non-Western repertoires. I would welcome the chance to learn from colleagues working on Chinese music history and performance, and I am committed to acquiring working Mandarin during the fellowship.

I attach my academic portfolio (CV, research statement, teaching statement, list of publications) and the five arXiv papers as my representative academic achievements from the past five years. I would be very grateful for the opportunity to discuss how my research and teaching trajectory might align with the team's current and planned activity. Further materials are available on request.

Thank you very much for your consideration.

Yours sincerely,

Dr Ignasi Solé Piñas

2. Curriculum Vitae

Dr Ignasi Solé Piñas

Musicologist · Ph.D. in Empirical Performance Studies

London, W13 0FD, UK · +44 7927 946 976 · ignasiphd@gmail.com · linkedin.com/in/isole

Profile

Musicologist working at the intersection of empirical performance studies, music historiography and music education. Doctoral research on tempo, performance time and portamento in Beethoven's piano and cello sonatas has yielded five arXiv preprints (2025–2026), a 346-page thesis, and an open dataset and analysis code in Python and MATLAB. The work spans technical method (a novel manual tempo-measurement protocol, k-means clustering of recorded traditions, a spline-CDF smoothing procedure for density estimation), historical critique (a quantified comparison of Czerny, Moscheles and Kolisch against twentieth-century performing practice), and a sociocultural account of how coexisting performance traditions persist and shift across decades. Ph.D. self-funded through over five years of senior music leadership in UK independent schools, including research-informed curriculum design from primary through pre-university level.

Education

- **Ph.D., Musicology** — University of Aberdeen (2021–2025). Supervisor: Dr Jonathan Hicks. Thesis: *Tempo and Performance Style in Recordings of Beethoven's Cello Sonatas, 1930–2012* (346 pp.).
- **Master's, Cello Performance** — Codarts University of the Arts, Rotterdam (2015–2017). With distinction.
- **Bachelor's, Cello Performance** — Liceu Conservatory of Barcelona (2011–2015).

Research Interests

- Empirical performance studies; historical recordings as evidence for stylistic change.
- Music historiography and criticism: the relationship between editorial authority and recorded practice.
- Sociocultural accounts of performance traditions; the transmission and revision of stylistic norms.
- Quantitative methods in musicology: tempo analysis, k-means clustering, data visualisation, spline-based density estimation.
- Music education informed by empirical research on performance.
- History of cello pedagogy; portamento, fingering, and the relationship between technique and expression.

Publications

Peer-reviewed preprints (arXiv).

- Solé, I. (2026). *Towards Revised Tempo Indications for Beethoven's Piano and Cello Sonatas: Czerny, Moscheles, Kolisch, and Recorded Practice 1930–2012.* arXiv:submit/7494591 [cs.SD].
- Solé, I. (2026). *A Complementary Visualisation Suite for Empirical Performance Analysis: Tempographs, Histograms, Ridgeline Plots, Stacked Bar Charts, and Combination Charts Applied to Beethoven's Piano and Cello Sonatas.* arXiv:submit/7494267 [cs.SD].
- Solé, I. (2026). *Coexisting Tempo Traditions in Beethoven's Piano and Cello Sonatas: A K-means Clustering Analysis of Recorded Performances, 1930–2012.* arXiv:submit/7492708 [cs.SD].
- Solé, I. (2026). *A Manual Bar-by-Bar Tempo Measurement Protocol for Polyphonic Chamber Music Recordings: Design, Validation, and Application to Beethoven's Piano and Cello Sonatas.*

arXiv:2604.15278 [cs.SD].

- Solé, I. (2025). *Evolving Performance Practices in Beethoven's Cello Sonatas: Tempo, Portamento, and Historical Interpretation of the First Movements.* arXiv:2502.00030 [cs.SD].

Doctoral thesis. Solé, I. (2025). *Tempo and Performance Style in Recordings of Beethoven's Cello Sonatas, 1930–2012.* Ph.D. dissertation, University of Aberdeen. 346 pp.

Open datasets and software. *PhD Appendix: Tempo Dataset and Coded Graphs.* Complete bar-level BPM data, annotated portamento scores, and Python/MATLAB analysis code. github.com/isolepinas/PhD-Appendix.

In preparation. Monograph proposal extending the doctoral argument to Beethoven's violin sonatas and selected string quartets (target 2026–2027); methods paper presenting the spline-CDF smoothing procedure as a standalone open-source tool; paper on Beethoven's historical pianos and their implications for tempo practice.

Research Experience

Independent Doctoral Researcher, University of Aberdeen **2021 – 2025**

Supervisor: Dr Jonathan Hicks

- Designed and executed a multi-year empirical research programme on tempo, performance time and portamento in Beethoven's five piano and cello sonatas, from research questions through methodology, data collection, analysis and writing.
- Assembled, digitised and annotated a research corpus of over 100 commercial and archival recordings spanning 1930–2012, including recordings by Casals, Feuermann, Fournier, Piatigorsky, du Pré, Rostropovich, Shafran, Starker, Tortelier, Ma, Wispelwey, Bylsma, Maisky, Perényi, Kliegel and Isserlis.
- Developed an original manual bar-by-bar tempo-measurement protocol after documenting the failure of automated beat-detection tools on polyphonic duo recordings; characterised the method to millisecond resolution.
- Implemented k-means clustering (scikit-learn, k-means++ initialisation, 100 random restarts) on the full BPM dataset, identifying three coexisting tempo traditions stable across eight decades.
- Developed a novel spline-CDF smoothing method for histogram density estimation as a contribution to the music-technology and performance-analysis toolkit.
- Produced a five-tool visualisation suite (tempographs, histograms with spline PDFs, ridgeline plots, stacked bar charts, combination charts) implemented in Python and MATLAB and openly published on GitHub.
- Developed a spectrographic gradient method for measuring portamento steepness in historical recordings, integrating Sonic Visualiser, GIMP image analysis and score annotation in Sibelius.

Cross-Disciplinary Collaboration, KTH Royal Institute of Technology **2021 – 2025**

Collaborated with Jordi Altayó (KTH, VLSI design) on the mathematical framework, error model and data architecture of the tempo-measurement protocol. Built an open-access reproducible research pipeline in Python (pandas, numpy, scipy, matplotlib, seaborn, scikit-learn) and MATLAB.

Teaching Experience

Notting Hill and Ealing High School (GDST) **Sep 2022 – Present**

Director of Music, Junior Department

- Design and deliver a research-informed seven-year music curriculum for 300+ students from Reception to Year 6, integrating performance, analysis, composition, digital music-making and music technology (BandLab, AI-assisted composition).

- Lead and mentor a team of 15+ visiting instrumental teachers; manage departmental budget, resource planning, safeguarding and assessment.
- Achieved a 60% Distinction/Merit rate in ABRSM exams through an in-house, low-stakes assessment model; grew the instrumental programme to 134 students across 12+ instruments.
- Established a flagship partnership with the Royal Opera House "Create and Dance" programme; directed performances at the Royal Albert Hall, Cadogan Hall and the GDST Sing festival.
- Mentor pre-university students preparing Oxbridge and conservatoire applications, including one-to-one supervision of analytical and historical writing.

St James Independent Prep School & The Falcons Pre-Prep School Sep 2020 – Aug 2022

Head of Music (concurrent roles)

- Led music provision across two independent schools spanning EYFS, KS1 and KS2; redesigned the Prep curriculum at St James around data-driven assessment.
- Reduced departmental expenditure at The Falcons by 50% over three years while maintaining quality of provision; mentored and appraised teaching staff across both schools.

Sistema Scotland & University of Aberdeen Aug 2017 – Aug 2020

Senior Music Educator, Strings Programme Lead & Director of Orchestral Music

- Delivered the Big Noise El Sistema programme to 600+ children weekly, Early Years to Year 9, with strings and vocal pedagogy rooted in ensemble work and social inclusion — direct experience of community-based arts practice.
- Directed the University of Aberdeen Symphony and Chamber Orchestras: weekly rehearsals, termly concerts and fundraising performances.
- Mentored undergraduate music-education placements in partnership with the University of Aberdeen School of Music.

Research Plans

Active projects in development: (i) a monograph proposal extending the doctoral argument from the cello sonatas to Beethoven's violin sonatas and selected string quartets, with a fuller historiographical treatment of the editorial tradition (Czerny, Moscheles, Kolisch) and the cultural history of portamento; (ii) a methods paper and standalone open-source release of the spline-CDF density-estimation procedure for use by other performance-corpus researchers; (iii) a hybrid manual-and-automated audio-analysis pipeline for polyphonic chamber recordings, designed to be adaptable beyond the Western canon and offering a possible point of contact with Chinese performance-research traditions. Near-term funding targets include the China Postdoctoral Science Foundation, NSSFC programmes appropriate to musicology and the humanities, and Shenzhen municipal postdoctoral schemes.

Awards, Scholarships and Funding

- Self-funded doctoral study, 2021–2025 — Ph.D. research entirely supported through concurrent full-time school leadership work; no external Ph.D. funding sought or received.
- Distinction, Master's Degree in Cello Performance, Codarts University of the Arts, Rotterdam, 2017.
- Partial scholarship, Codarts University of the Arts, Rotterdam, 2015–2017.
- Entrance scholarship, Liceu Conservatory of Barcelona, 2011.

Academic Service and Outreach

- Maintainer of the open-access *PhD Appendix* repository on GitHub, providing the complete tempo dataset and analysis code for reuse by other researchers.

- Public engagement: curriculum talks for sixth-form and pre-university music students on research methods in musicology and on the historical recording tradition.
- Peer correspondence with practising performers (cellists and pianists) including Steven Isserlis, on the implications of empirical performance research for contemporary interpretation.

Professional Memberships and Qualifications

- National Professional Qualification for Senior Leadership (NPQSL).
- Qualified Teacher Status (QTS), Teacher Reference Number 4070040.
- Designated Safeguarding Lead, Level 3.
- Full Paediatric First Aid (current).

Technical Skills

Research software. Audio analysis: Sonic Visualiser, Audacity, Spleeter. Programming: Python (pandas, numpy, scipy, matplotlib, seaborn, scikit-learn), MATLAB, working knowledge of R and Stata. Data structuring and visualisation: Jupyter notebooks, LaTeX, Google Sheets, Microsoft Excel. Score preparation: Sibelius, Dorico, MuseScore. Image and spectrogram analysis: GIMP. Version control and open science: Git, GitHub, arXiv submission workflows.

Languages. Catalan and Spanish: native. English: fluent (full professional proficiency). Italian: intermediate reading, basic spoken. Mandarin: beginner — committed to acquiring working proficiency during the fellowship. Swedish: beginner.

Referees

Available on request. Principal Ph.D. supervisor Dr Jonathan Hicks (University of Aberdeen) and two additional referees can be contacted on request.

3. Research Statement

Track record

My research lies in empirical performance studies, a field that treats recordings as primary evidence for how musical works have been interpreted and for how interpretation has changed over time. My doctoral thesis, submitted to the University of Aberdeen in 2025, is the first systematic empirical study of Beethoven's five sonatas for piano and cello across the full eight decades of their recorded history. It is a 346-page contribution to a body of work that includes José Bowen's and Nicholas Cook's foundational studies of tempo and expressive flexibility, Daniel Leech-Wilkinson's portamento research, Marten Noorduin's historiographical account of Beethoven's tempo indications, and the broader CHARM tradition.

Three substantive contributions have emerged from this research and now form the basis of five papers on arXiv and of a planned monograph.

First, a methodological contribution. I developed a formalised manual bar-by-bar tempo-measurement protocol for polyphonic chamber recordings, in response to the systematic failure of automated beat-detection tools on historical duo material. The protocol rests on a cumulative timestamp architecture that prevents error accumulation, permits internal self-validation, and captures the expressive timing phenomena (rubato, fermatas, ritardandi) that automated tools suppress. It was developed in cross-disciplinary collaboration with an engineer specialising in VLSI design, and its error model, data architecture and quality-control procedures are characterised in full in my 2026 arXiv paper on the protocol. The complete dataset and analysis code are publicly available, and the protocol has been designed from the outset to be adoptable and extendable by other researchers working with similar material.

Second, a substantive finding about stylistic change. Applying k-means clustering to the bar-level BPM data, I have shown that every movement of the five sonatas supports at least two and usually three discrete tempo traditions (labelled slow, mid-range and fast) whose internal regression slopes are negligible across the entire 1930–2012 period. The dominant mid-range tradition has been internally stable across eight decades; what has changed at the aggregate level is a shift in the relative prevalence of coexisting traditions, not a uniform drift of all performers toward a new norm. This finding supports what I have called an *ecological model of stylistic change*: coexisting traditions fluctuate in relative density, rather than a single tradition evolving. It reframes how corpus-level regression results in earlier performance studies should be interpreted, and it has clear implications for how future empirical studies should report their data.

Third, a finding about Beethoven's historical tempo indications. My comparative analysis of Czerny's, Moscheles's and Kolisch's markings against the recorded corpus documents systematic gaps ranging from 11–20% in fast movements to 37–39% in some slow movements. Kolisch's 1943 markings, derived without corpus evidence, align considerably more closely with recorded practice than either Czerny's or Moscheles's — a striking result that suggests his musical intuitions tracked something the broader tradition had not yet been able to measure. From this comparison I propose a set of revised tempo indications grounded in the statistical modal tempi of the corpus, presented as ranges rather than single prescriptive values. These are offered not as claims about Beethoven's intentions but as evidence-based reference points for performers and scholars navigating the gap between historical prescription and performable reality.

Future plans

My immediate research priority is the publication of the doctoral work in book form. I plan to submit a monograph proposal to a major university press within the next year, extending the cello-sonatas argument into adjacent repertoires and deepening the historiographical dimensions that space constraints in the thesis did not permit me to explore fully. The monograph will incorporate the five arXiv papers as revised chapters and add substantial new material on Beethoven's historical pianos, on the cultural history of portamento, and on the broader implications of the ecological model of stylistic change for performance studies as a discipline.

In parallel, I intend to extend the methodology to the Beethoven violin sonatas and selected string quartets. The violin sonatas are an obvious test case because their recorded tradition is richer than the cello sonatas' and because violin technique and portamento history diverge from cello practice in instructive ways. A full k-means clustering analysis of the violin sonatas would test whether the three-tradition structure I documented in the cello works generalises across Beethoven's chamber output, or whether it is specific to the cello-piano duo. Preliminary work on this project can begin immediately, and I anticipate one or two papers emerging from it within two years.

A third direction is methodological. My spline-CDF smoothing procedure for histogram density estimation is a small but principled contribution that I would like to extend. The method is specifically suited to discrete, bounded performance data and addresses artefacts that standard kernel density estimation produces at repeated values and at distribution boundaries. I plan to develop the method into a standalone tool, published with a short methods paper, that other researchers can adopt for performance corpus analysis without needing to implement it from scratch. A related project is a full open-source implementation of the five-tool visualisation suite as a documented Python package.

A fourth direction, which I would welcome the opportunity to develop in dialogue with colleagues at Shenzhen, is a hybrid manual-and-automated audio-analysis pipeline for polyphonic chamber recordings. The manual protocol I have developed is labour-intensive but yields ground-truth data of a quality that is hard to obtain at scale. A hybrid approach, in which the manual protocol is used to generate training and validation data for more targeted automated tools, would benefit both empirical musicology and music-information retrieval: it would give MIR researchers a principled way to validate their models on historical material, and it would give musicologists a path to scalability that manual work cannot offer on its own. The pipeline would be designed to be adaptable beyond the Western canon, and Shenzhen — with its strong cross-disciplinary research culture and its position within the wider Chinese musicological community — is an ideal environment in which to develop that adaptability.

Finally, I am committed to the public dimension of this research. My open-access dataset and code are already freely available, and I would like to continue developing resources that make empirical performance analysis more accessible to performers, teachers, and students at every level. The gap between the tempo indications that students are taught to respect and the tempi that expert performers have actually adopted across the entire recording era is one of the most striking findings of my thesis, and it has immediate pedagogical implications.

The combination of a substantial completed research programme, a clear publication pipeline, and concrete plans for extension gives me a research agenda that I believe would fit well with the work of Professor Liu Kun's team. Three points of contact are particularly direct: the historiographical and music-criticism dimensions of the Czerny/Moscheles/Kolisch paper; the sociocultural framing of the ecological model of stylistic change; and the music-education resonances of the recorded-practice findings. I would welcome the chance to discuss how that agenda might align with current and planned research activity at Shenzhen.

4. Teaching Statement

Teaching experience

I have taught continuously since 2017, first as a practical instrumental teacher and chamber-music coach, then as a classroom teacher of music, and since 2022 as Director of Music at Notting Hill and Ealing High School in London, a role in which I design and lead the music curriculum for more than three hundred students ranging from Year 3 through Year 13. While my classroom teaching has been at primary and secondary level rather than at university, I have consistently operated at a level that bridges school and tertiary pedagogy: I have mentored Year 12 and 13 students through Oxbridge and conservatoire applications, supervised individual research projects on performance and historical topics, and prepared bespoke teaching resources for pre-university students on analytical and historical writing about music.

My approach to teaching has been shaped by two convictions. The first is that each student can be drawn into engaged intellectual work if the teacher is willing to meet them where they are and then move with them toward where they could go. The second is that analytical thinking and creative exploration are not opposed activities but mutually reinforcing ones: students who have been taught to listen carefully and to ask precise questions about what they hear make more adventurous performers and composers, and students with active creative practices bring an interpretative sharpness to analytical work that would otherwise take years to cultivate. These convictions have informed the curriculum I design, which integrates performance, composition, listening and written analysis from the earliest years rather than treating them as separate specialisations to be chosen between.

Teaching philosophy

I teach with a strong preference for evidence and example. In my own research I have argued that aggregate claims about performance practice can be misleading when the underlying population is plural, and I try to apply the same scrupulousness to teaching. When I introduce a concept (rubato, historically informed performance, the classical sonata), I prefer to do it through contact with the primary material: students listen to specific recordings, look at specific scores, and make their own observations before any received framework is offered. The framework then arrives as a way of organising what they have already heard, rather than as an authority to be accepted on trust. This approach is slower than a lecture-led style but consistently produces the deeper and more durable kind of understanding.

I also believe firmly in modelling uncertainty and disagreement. Musicology, and especially empirical performance studies, is a discipline in which reasonable scholars disagree about what the evidence shows and about how to weigh competing sources. Teaching well means making that disagreement visible rather than hiding it behind a textbook consensus. In my classroom and small-group work I try to show students how a scholar builds an argument, where the weak points are, and how an alternative view would be constructed. Students respond well to this, and it prepares them for the kind of independent critical work that university study requires.

Given my research background and teaching experience, I am prepared to design and deliver undergraduate and postgraduate modules on a range of topics, including: empirical methods in performance studies; history of cello performance and chamber music; historically informed performance; Beethoven's chamber music; introductory quantitative methods for musicologists (tempo analysis, basic statistics, data visualisation); score and audio analysis; and academic writing for music students. I would also be prepared to co-teach modules on music technology and recording history, and to contribute to supervision of undergraduate and Master's dissertations on topics in performance studies and recording analysis.

At postgraduate level, my research methodology is well suited to supporting doctoral students whose projects require large-corpus recording analysis, empirical testing of historical performance claims, or hybrid manual-and-automated audio analysis workflows. I would be particularly keen to co-supervise projects that bridge musicology and music-information retrieval, an interdisciplinary space in which my manual protocol provides a useful methodological starting point.

I am candid about the fact that my classroom teaching has not been at university level, and I recognise that university teaching makes different demands: larger lectures, specialised seminars, supervision at dissertation and thesis level, and close integration of teaching with active research. I believe my track record demonstrates that I am equipped to make this transition well. My five years of senior music leadership in demanding academic schools have given me substantial experience of curriculum design, teaching team management, and sustained communication with students, colleagues and parents. My doctoral work, produced in parallel with full-time school leadership, demonstrates the research capacity and the self-discipline that university roles require. And my continuing mentoring of pre-university students gives me a concrete, if small-scale, experience of supervising individual analytical projects at a level close to first-year undergraduate work.

I would welcome a structured induction into university-level teaching in whatever form the department finds useful. I am keen to develop as a university educator, and I view that development as a natural continuation of a trajectory that has already taken me from performance to teaching to research, each stage informed by the last.

5. List of Publications

Doctoral thesis

Solé, I. (2025). *Tempo and Performance Style in Recordings of Beethoven's Cello Sonatas, 1930–2012.* Ph.D. dissertation, University of Aberdeen. 346 pp. Supervisor: Dr Jonathan Hicks.

Abstract. This study examines the effect of time on the performance practices of Ludwig van Beethoven's sonatas for piano and cello, focusing specifically on tempo, performance time and portamento. Inspired by research from José Bowen, George Kennaway, Daniel Leech-Wilkinson, Clive Brown and Marten Noorduin, the thesis provides a detailed analysis of sonatas that have not been previously examined in this manner. Findings reveal a notable increase in tempo and a decline in portamento usage since the 1970s, suggesting a distinct shift in interpretative style. Drawing on data from over a hundred recordings by influential cellists of the twentieth and twenty-first centuries, the research highlights how cellists have adapted their tempi and fingerings, often making portamento redundant and thereby aligning with evolving stylistic preferences that prioritise clarity and agility. The analysis considers additional factors such as age and regional performance traditions; although these factors offer valuable context, findings indicate that the evolution in performance practices is ultimately driven by individual artistic interpretation and collaborative decisions between cellists and pianists. Through a comparative approach anchored between the 1930 Casals–Horszowski and the 2012 Isserlis–Levin recordings, the research offers realistic tempo recommendations aimed at informing future interpretations.

Peer-reviewed preprints (arXiv)

1. Solé, I. (2026). *Towards Revised Tempo Indications for Beethoven's Piano and Cello Sonatas: Czerny, Moscheles, Kolisch, and Recorded Practice 1930–2012.* arXiv:submit/7494591 [cs.SD], 18 April 2026.
   Presents the first systematic empirical assessment of Czerny's, Moscheles's and Kolisch's historical metronome indications against a corpus of over one hundred movement-level recordings. Documents systematic gaps averaging 11–20% in fast movements and 37–39% in some slow movements; shows that Kolisch's 1943 markings align more closely with recorded practice than either Czerny's or Moscheles's. Proposes revised tempo indications as ranges grounded in the statistical modal tempi of the recorded corpus.

2. Solé, I. (2026). *A Complementary Visualisation Suite for Empirical Performance Analysis: Tempographs, Histograms, Ridgeline Plots, Stacked Bar Charts, and Combination Charts Applied to Beethoven's Piano and Cello Sonatas.* arXiv:submit/7494267 [cs.SD], 18 April 2026.
   Presents a suite of five complementary visualisation tools for bar-level tempo data and argues that reliance on any single type systematically conceals features the others expose. Introduces a spline-CDF smoothing method for histogram probability density functions as a novel contribution to the performance-analysis toolkit. A five-panel composite figure demonstrates that five different projections of the same pair of recordings (Casals/Horszowski 1930 and Isserlis/Levin 2012) yield different and complementary findings.

3. Solé, I. (2026). *Coexisting Tempo Traditions in Beethoven's Piano and Cello Sonatas: A K-means Clustering Analysis of Recorded Performances, 1930–2012.* arXiv:submit/7492708 [cs.SD], 17 April 2026.
   Applies k-means clustering ($k = 3$, k-means++ initialisation, 100 random restarts) to bar-level BPM data from 100+ movement-level recordings. Reveals that every movement supports at least two and usually three discrete tempo traditions whose internal regression slopes are negligible. Proposes an ecological model of stylistic change in which coexisting traditions shift in relative prevalence rather than a single tradition evolving. Finds no correlation between cluster membership and performers' generational, national or pedagogical backgrounds.

4. Solé, I. (2026). *A Manual Bar-by-Bar Tempo Measurement Protocol for Polyphonic Chamber*

Music Recordings: Design, Validation, and Application to Beethoven's Piano and Cello Sonatas. arXiv:2604.15278 [cs.SD], 16 April 2026.
Documents the systematic failure of automated beat-detection tools on historical duo recordings and presents a formalised manual alternative. The protocol rests on a cumulative timestamp architecture that prevents error accumulation, permits internal self-validation, and captures expressive timing phenomena that automated tools suppress or misread. The mathematical derivation, spreadsheet data structure and error characterisation are presented in full. Developed in cross-disciplinary collaboration with an engineer specialising in VLSI design.

5. Solé, I. (2025). *Evolving Performance Practices in Beethoven's Cello Sonatas: Tempo, Portamento, and Historical Interpretation of the First Movements.* arXiv:2502.00030 [cs.SD], 23 January 2025.
The first paper of the series. Integrates analyses of 22 historical recordings, examines Beethoven's metronome markings as understood through Czerny and Moscheles, and documents the diminishing use of audible portamento in the latter half of the twentieth century alongside a gradual increase in tempo after 1970. Introduces the concept of "silent portamento" as a new analytical variable for studies of expression in string performance.

Open datasets and software

Solé, I. (2024–2026). *PhD Appendix: Tempo Dataset and Coded Graphs.* GitHub repository, github.com/isolepinas/PhD-Appendix. Includes the complete bar-level BPM dataset for all analysed recordings, annotated portamento scores, and Python/MATLAB analysis code covering all five visualisation types and the k-means clustering pipeline.

In preparation

- A monograph based on the doctoral thesis, extending the argument to selected works from Beethoven's violin sonatas and string quartets. Target submission: 2026–2027.
- A paper on Beethoven's historical pianos and their implications for tempo practice, drawing on Chapter 2 of the thesis.
- A methods paper presenting the spline-CDF smoothing procedure as a standalone open-source tool for performance-analysis researchers.